Table 1. Predictive relevance of potential experience instruments

Baseline experience specification

Panel A. NLSY79

Dependent variable	E	E^2	E^3	$S \times E$	$A \times E$
Joint F statistic	81.60	29.87	17.90	81.05	39.72
Joint test p value	<0.001	<0.001	<0.001	<0.001	<0.001
Partial R^2	0.223	0.165	0.097	0.253	0.393
R^2	0.878	0.738	0.502	0.876	0.906
Observations	30,246	30,246	30,246	30,246	30,246
Respondents	7,631	7,631	7,631	7,631	7,631
Excluded instruments	5	5	5	5	5

Panel B. NLSY97

Dependent variable	E	E^2	E^3	$S \times E$	$A \times E$
Joint F statistic	138.85	61.84	39.18	146.36	47.24
Joint test p value	<0.001	<0.001	<0.001	<0.001	<0.001
Partial R^2	0.332	0.356	0.366	0.368	0.519
R^2	0.878	0.892	0.879	0.883	0.940
Observations	11,438	11,438	11,438	11,438	11,438
Respondents	2,889	2,889	2,889	2,889	2,889
Excluded instruments	5	5	5	5	5

Notes. E denotes actual experience, P potential experience, S schooling, and A the ASVAB score. Each column is a separate regression containing all five excluded instruments: P, P squared, P cubed, S times P, and A times P.

Every regression includes schooling, ASVAB, tenure through its cubic term, schooling times tenure, ASVAB times tenure, and indicators for year, sex, race, and original job number. Schooling, ASVAB, and tenure terms are treated as exogenous. Sampling weights are used and inference is clustered by respondent.

The sample retains original job numbers 1 to 3, NLSY79 observations through 2004, and all available NLSY97 years. Eligibility follows the existing pooled wage winsorization at the 1st and 99th percentiles. P is survey year less earliest recorded employment year; it includes nonemployment after entry. It is an observed entry proxy and may include student work. NLSY79 starts coded as week zero have unknown dates before 1978 and are left missing.

Requiring usable potential experience reduces the eligible NLSY79 sample from 44,532 to 30,246 observations, and NLSY97 from 11,440 to 11,438. Partial R squared measures the additional explanatory power of the full excluded instrument set, conditional on the included controls. Joint F tests have 5 numerator degrees of freedom and the number of respondents less one denominator degrees of freedom. These tests do not establish exclusion validity or conditional strength for the joint IV system.

Table 2. First stage coefficients on excluded instruments

Coefficients with respondent clustered standard errors in parentheses

Panel A. NLSY79

Excluded instrument	E	E^2	E^3	$S \times E$	$A \times E$
P	−0.0630 (0.1961)	−6.4439 (5.7907)	18.5543 (177.4009)	−7.0044** (2.8594)	0.4480 (0.4531)
P^2	0.0312*** (0.0071)	0.7560*** (0.2250)	−5.2071 (8.8152)	0.3863*** (0.0937)	0.0540*** (0.0171)
P^3	−0.0006** (0.0002)	0.0033 (0.0088)	0.9714** (0.3810)	−0.0072** (0.0031)	−0.0016*** (0.0005)
$S \times P$	0.0187 (0.0177)	0.1579 (0.5189)	−2.9134 (15.1641)	0.7589*** (0.2605)	−0.0462 (0.0409)
$A \times P$	0.0505*** (0.0163)	1.5157** (0.6751)	45.9944* (25.3627)	0.6368*** (0.2209)	0.5890*** (0.0604)
Joint F statistic	81.60	29.87	17.90	81.05	39.72
Partial R^2	0.223	0.165	0.097	0.253	0.393
Observations	30,246	30,246	30,246	30,246	30,246
Respondents	7,631	7,631	7,631	7,631	7,631

Panel B. NLSY97

Excluded instrument	E	E^2	E^3	$S \times E$	$A \times E$
P	0.2143 (0.1715)	−9.4866** (4.6595)	−195.4017* (116.3237)	−5.6471* (2.9718)	−0.1136 (0.3907)
P^2	0.0300*** (0.0076)	1.0459*** (0.2057)	5.7715 (6.1718)	0.3677*** (0.1015)	0.0282 (0.0201)
P^3	−0.0006*** (0.0002)	−0.0068 (0.0069)	0.5378** (0.2202)	−0.0075** (0.0030)	−0.0009 (0.0006)
$S \times P$	0.0101 (0.0118)	0.3451 (0.3262)	9.2768 (8.1054)	0.8096*** (0.2086)	0.0100 (0.0255)
$A \times P$	0.0108 (0.0121)	0.0086 (0.3025)	−3.3578 (6.9613)	0.1159 (0.1742)	0.6142*** (0.0467)
Joint F statistic	138.85	61.84	39.18	146.36	47.24
Partial R^2	0.332	0.356	0.366	0.368	0.519
Observations	11,438	11,438	11,438	11,438	11,438
Respondents	2,889	2,889	2,889	2,889	2,889

Notes. E denotes actual experience, P potential experience, S schooling, and A the ASVAB score. Each column is a separate regression containing all five excluded instruments: P, P squared, P cubed, S times P, and A times P.

Estimates are reported in original units. Standard errors appear beneath coefficients. *** p < 0.01, ** p < 0.05, * p < 0.10. These stars concern individual coefficients, whereas the joint F statistic tests all five excluded instruments together.

Every regression includes schooling, ASVAB, tenure through its cubic term, schooling times tenure, ASVAB times tenure, and indicators for year, sex, race, and original job number. Schooling, ASVAB, and tenure terms are treated as exogenous. Sampling weights are used and inference is clustered by respondent.

The samples and entry definition are identical to Table 1. Control coefficients are included in estimation and omitted from the display. No structural wage coefficients are estimated in these tables.

Table 3. Occupation history prediction checks

Supplementary evidence on predictive relevance

Panel A. NLSY79

Current occupation	History F	History p	Full set F	Full set partial R^2
Professional	63.27	<0.001	59.56	0.422
Managers	34.13	<0.001	33.30	0.347
Sales	11.50	<0.001	11.09	0.227
Clerical	76.76	<0.001	71.02	0.383
Craftsmen	36.54	<0.001	35.37	0.326
Operatives and laborers	59.82	<0.001	55.77	0.359

Observations: 15,089; respondents: 3,746; history indicators tested: 48; total excluded instruments: 53.

Panel B. NLSY97

Current occupation	History F	History p	Full set F	Full set partial R^2
Professional	152.97	<0.001	142.63	0.711
Managers	1,384.36	<0.001	1,354.73	0.653
Sales	5,081.37	<0.001	5,401.60	0.709
Clerical	97.03	<0.001	91.05	0.692
Craftsmen	3,420.30	<0.001	3,610.77	0.802
Operatives and laborers	318.29	<0.001	371.14	0.741

Observations: 4,555; respondents: 1,111; history indicators tested: 47; total excluded instruments: 52.

Notes. Each row is a linear probability regression for the indicated current occupation. Service is the omitted outcome category. Each regression includes the five potential experience terms and the same exogenous controls as Table 1, together with interactions between occupation indicators from the previous two survey waves. Only current main job rows with both lagged occupations observed are included. Histories use the previous two scheduled survey waves, including biennial waves; missing interviews are not skipped.

One observed history pair is omitted with the intercept. Empty history pairs are excluded. History F tests only the occupation history block, conditional on the potential experience instruments and included controls. Full set F and partial R squared refer to all excluded instruments together. Standard errors are clustered by respondent and sampling weights are used. Joint test p values below 0.001 are displayed as <0.001.

These are supplementary prediction checks. They do not implement the full transformed wage model or nonlinear IV estimation in Gibbons et al. (2005). The occupation interactions required by the full thesis wage model are not tested here, and relevance alone does not establish instrument validity.

Presentation reference: Gibbons, Katz, Lemieux, and Parent (2005), Journal of Labor Economics, 23(4), Table B3. DOI: 10.1086/491606.
Formatting assistance: Codex; numerical estimates are read directly from the saved Stata exports.